# Doubles for both triple reconstruction and assay validation

torchcell lab notes

**Script** `experiments/010-kuzmin-tmi/scripts/construction_validation_doubles.py`
**Data** `experiments/010-kuzmin-tmi/results/construction_validation_doubles.csv`
**Dendron** `[[experiments.010-kuzmin-tmi.scripts.construction_validation_doubles]]`

---

## Contents

# 1 Why the pure set-cover was the wrong objective

The essential wet-lab task is a small set of double mutants that both (a) reconstructs many top-ranked triples and (b) has **variance in DMF / interaction** to validate the new echo-plating assay against (versus plate-stamping). The triple-coverage set-cover (`[[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]`) does (a) with 8 doubles but fails (b): those 8 are all **near-neutral** — Costanzo DMF span 0.15, $|\varepsilon| \leq 0.039$, and **zero significant interactions** — so the assay would have almost nothing to discriminate. An assay validated only on near-1.0 fitness and null interactions has not been shown to detect signal.

# 2 Two tiers, unioned (13 doubles)

*coverage (8)* the greedy set-cover doubles; reconstruct **all 31** within-10 top-$k$ triples.
*validation (5)* added for dynamic range and real signal: the doubles with a **significant Costanzo interaction** ($P < 0.05$ and $|\varepsilon| > 0.08$, both signs), the **highest-DMF** double, and a **low-DMF anchor chosen for quality** — among low-DMF ($< 0.75$) doubles, the one that also reconstructs triples with the tightest SD (YER079W+YJR060W: DMF $0.610 \pm \mathbf{0.018}$, covers 3 triples), *not* the noisy DMF-minimum (YGL087C+YJR060W was $0.513 \pm \mathbf{0.172}$).
*novel (1, separate)* **YPL046C+YPL081W (ELC1+RPS9A)**, the one within-10 pair with **no DMF in Costanzo or Kuzmin** — a construction candidate to fill the last cell (Section 6).

Result: 31/31 triples still reconstructed, DMF range **0.610–1.178**, $\varepsilon$ from $-\mathbf{0.130}$ to $+\mathbf{0.098}$, **3 significant interactions**.

| # | Double | Tier | DMF ± SD | $\varepsilon$ | $p$ | sig | triples |
|---|---|---|---|---|---|---|---|
| 1 | YER079W+YJR060W | validation | $0.610 \pm 0.018$ | $-0.003$ | 0.454 | | 3 |
| 2 | YER079W+YPL081W | validation | $0.863 \pm 0.098$ | $-0.130$ | 0.035 | ✓ | 0 |
| 3 | YJR060W+YKL033W-A | validation | $0.871 \pm 0.023$ | $-0.082$ | 0.007 | ✓ | 0 |
| 4 | YJR060W+YPL046C | coverage | $0.966 \pm 0.018$ | $+0.003$ | 0.453 | | 4 |
| 5 | YDR057W+YLL012W | coverage | $1.000 \pm 0.069$ | $-0.036$ | 0.296 | | 5 |
| 6 | YLR312C-B+YPL081W | coverage | $1.016 \pm 0.042$ | $-0.019$ | 0.355 | | 3 |
| 7 | YLL012W+YPL046C | coverage | $1.025 \pm 0.053$ | $-0.010$ | 0.429 | | 5 |
| 8 | YDR057W+YPL081W | coverage | $1.035 \pm 0.045$ | $+0.039$ | 0.260 | | 3 |
| 9 | YDR057W+YER079W | coverage | $1.096 \pm 0.054$ | $+0.012$ | 0.435 | | 5 |
| 10 | YER079W+YLR312C-B | coverage | $1.099 \pm 0.039$ | $-0.027$ | 0.290 | | 5 |
| 11 | YBR203W+YPL046C | coverage | $1.118 \pm 0.017$ | $+0.036$ | 0.077 | | 5 |
| 12 | YDR057W+YGL087C | validation | $1.137 \pm 0.024$ | $+0.098$ | 0.001 | ✓ | 3 |
| 13 | YDR057W+YLR312C-B | validation | $1.178 \pm 0.026$ | $+0.047$ | 0.119 | | 1 |
| 14 | YPL046C+YPL081W | novel | — | — | — | | 0 |

**Fig. 1** The 13 selected doubles by tier, with Costanzo DMF, SD, $\varepsilon$ and $p$.

# 3 Single mutants to inoculate (10)

To build the doubles above, cross these 10 single-mutant strains pairwise. **Every** panel-10 gene appears in at least one double, so all 10 are required — no smaller set builds them all. The `# doubles` column is how many each gene crosses into (YOS9/YDR057W is the busiest hub at 5; COS111, MMS2 and YKL033W-A each appear once). The novel pair adds no new strain (ELC1 and RPS9A are already in the set).

All 44 measured within-10 doubles ranked by DMF, coloured by tier (coverage, validation, unused; significant $\varepsilon$ ringed) — the tiered companion to `constructed_10_dmf_forest`, which predates the validation tier and marks only the 8 set-cover doubles. The 45th pair, YPL046C+YPL081W, is unmeasured and so absent here (Section 6).

| # | Systematic | Common | # doubles |
|---|---|---|---|
| 1 | YBR203W | COS111 | 1 |
| 2 | YDR057W | YOS9 | 5 |
| 3 | YER079W | — | 4 |
| 4 | YGL087C | MMS2 | 1 |
| 5 | YJR060W | CBF1 | 3 |
| 6 | YKL033W-A | — | 1 |
| 7 | YLL012W | YEH1 | 2 |
| 8 | YLR312C-B | (SPH1 locus) | 3 |
| 9 | YPL046C | ELC1 | 4 |
| 10 | YPL081W | RPS9A | 4 |

**Fig. 2** The 10 single-mutant strains to inoculate, with the number of selected doubles each participates in.

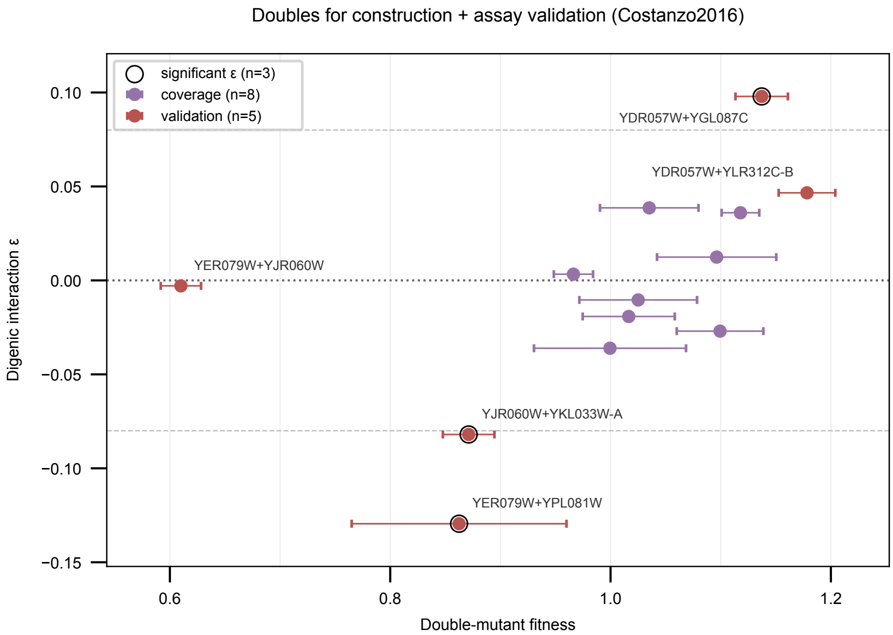


**Fig. 3** The selected doubles in interaction-versus-fitness space: Costanzo digenic interaction $\varepsilon$ against double-mutant fitness, with the coverage and validation tiers distinguished and the three significant interactions ringed. The validation tier is what spreads the set along both axes; the coverage tier alone clusters near $\varepsilon = 0$.

DMF, SD, SE, $\varepsilon$ and $p$ are the published Costanzo values (SD = sample SD over 4 colonies, directly comparable to the assay's colony SD — see `[[experiments.010-kuzmin-tmi.scripts.constructed_10_dmf_reference]]`). The low-DMF end is dominated by CBF1/YJR060W (its single defect), so a low DMF there is a fitness effect, not an interaction. Two of the three significant doubles (YER079W+YPL081W, YJR060W+YKL033W-A) enable no top-$k$ triple — they are pure validation targets; the low anchor YER079W+YJR060W and the strongest-positive-$\varepsilon$ YDR057W+YGL087C each also rebuild triples. The set is tunable: drop the two triple-less validation doubles for an 11-double set, or add more near-significant $\varepsilon$.

# 4 There are only 3 significant interactions in the whole panel

Surveyed across all 66 panel-12 doubles and all three sources: exactly **3** clear Costanzo's bar ($P < 0.05$ and $|\varepsilon| > 0.08$), and all 3 are within-10 and already in the validation tier — YDR057W+YGL087C (+0.098), YJR060W+YKL033W-A (−0.082), YER079W+YPL081W (−0.130). None hide in the dropped-gene doubles; Kuzmin 2018/2020 have too few measured pairs to reach significance. So "construct the significant ones" means these 3, already selected. The panel is near-neutral by design, so strong interactions are scarce.

## 4.1 Why a big interaction can still be "not significant" — and why that is fixable

A double's $\varepsilon$ is a point estimate; whether it counts as significant depends on how big it is *relative to its error bar*. The error bar on $\varepsilon$ is driven by the colony-to-colony scatter of the double *and* of the two singles it is compared against (all

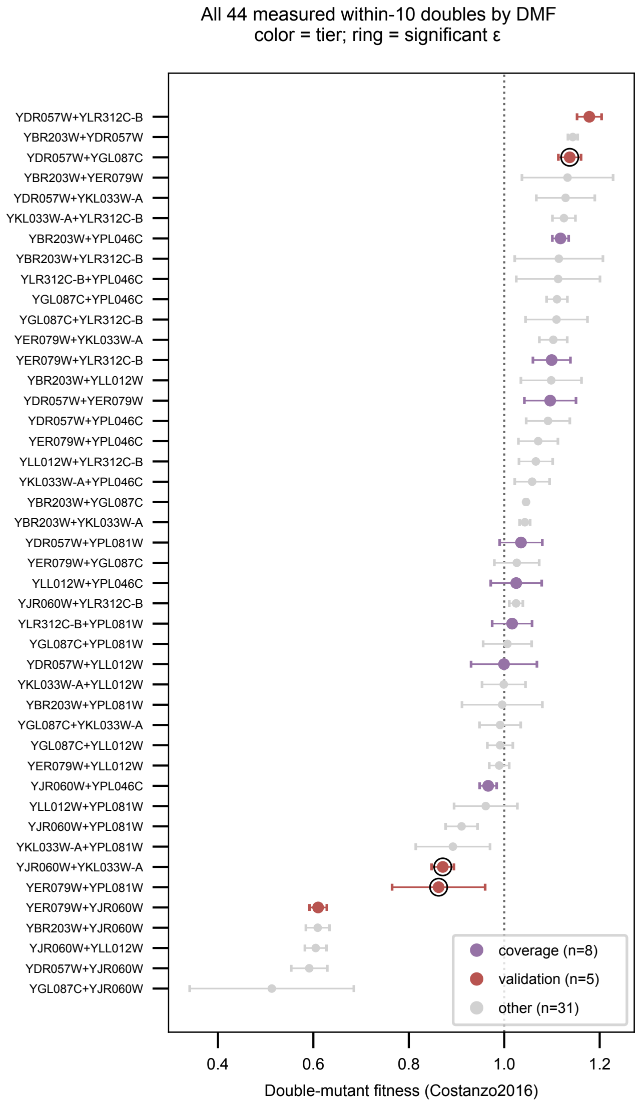


**Fig. 4** Forest plot of all 44 measured within-10 doubles, ranked by Costanzo DMF and coloured by tier.

$n = 4$ colonies in Costanzo). So a double sitting on a noisy strain (CBF1, RPS9A) can show a sizable $\varepsilon$ that is still inside its own error bar — insignificant. The reverse also happens: YBR203W+YDR057W has a tiny $\varepsilon$ (+0.062) that *is* significant ($p = 1.4 \times 10^{-5}$) only because its colonies were very tight (SD 0.010).

The key point for us: **that error bar is set by how many colonies we plate, and that is our choice.** More colonies gives a smaller error bar, shrinking with the square root of the count. So a "noisy but interesting" double is not disqualified — it just needs to be plated deeper. Our goal is depth (measure a few predicted doubles precisely), not breadth (screen many strains shallowly), which is exactly what echo dispensing is good at. So we pick the validation doubles for *interesting values* (fitness extremes, large $\varepsilon$) and drive each to a tight measurement at the bench.

One caveat: deeper plating tightens *our* error bar, but Costanzo's stays at $n = 4$. Once ours is tighter, a disagreement may be the reference's noise, not ours.

# 5 Cross-dataset check — Costanzo versus Kuzmin DMF (a real caveat)

The DMF values above are **Costanzo only**. Where Kuzmin also measured a within-10 double, the two references sometimes **disagree sharply**, especially for the low-DMF CBF1 doubles:

CBF1/YJR060W's low DMF is largely a *Costanzo* phenomenon — Kuzmin reads those pairs higher, and YJR060W+YLL012W disagrees by about 0.5. So the low-DMF "signal" our assay would validate against is reference-dependent; treat the low-DMF anchor's target loosely and, where a Kuzmin value exists, report both. The full per-double Costanzo+Kuzmin comparison is in the supplementary table (Section 7).

**Table 1** Within-10 doubles where Kuzmin also measured a DMF. The two references disagree sharply on the low-DMF CBF1 pairs – YJR060W+YLL012W by about 0.5 – so the low-DMF target the assay validates against is reference-dependent.

| double | Costanzo DMF | Kuzmin DMF |
|---|---|---|
| YJR060W+YLL012W | 0.605 | **1.119** (K2020) |
| YJR060W+YPL046C | 0.966 | 0.738 (K2018) |
| YBR203W+YJR060W | 0.609 | 0.768 (K2018) |
| YGL087C+YPL046C | 1.111 | 0.998 (K2018) |

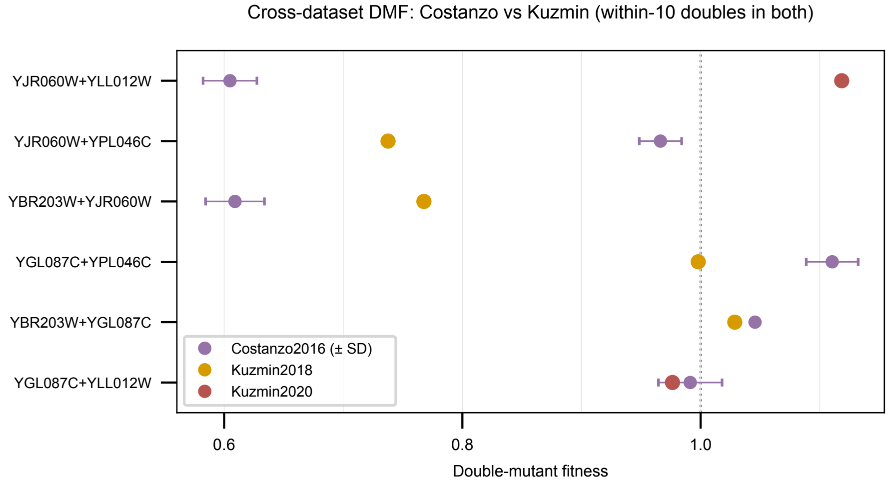


**Fig. 5** Dumbbell of every within-10 double measured in both references: Costanzo with SD, joined to the Kuzmin value.

### 5.1 The same ambiguity exists *within* Costanzo, at the strain level

The disagreement above is between *datasets*. There is a second, easier-to-miss ambiguity *inside* Costanzo: a gene can carry more than one strain, with materially different measurements. YJR060W (CBF1) is the case that matters for this panel —

**Table 2** The same gene, two Costanzo strains, two published single-mutant fitnesses 0.33 apart. The queried reference table picked `sn154` by a dictionary collision rather than by any stated rule, and CBF1 anchors the low end of every fitness correlation reported here.

| Costanzo strain | SMF | stddev |
|---|---|---|
| `YJR060W_dma2646` | 0.9230 | 0.0102 |
| `YJR060W_sn154` | 0.5900 | 0.1138 |

— two published single-mutant values 0.33 fitness units apart. The queried reference table the assay compares against picked `sn154` by a dictionary collision in `12_panel_inference_3_queried_data_tables.py` (the gene-keyed index is overwritten by each successive record, so the surviving row is whichever LMDB iteration reached last), not by any stated rule. The same code applies no temperature filter, so 26 °C versus 30 °C is also last-record-wins — moot for this panel, where all 12 ORFs happen to carry identical 26/30 entries, but not moot in general (17% of Costanzo strains differ between the two temperatures).

This matters because CBF1 anchors the low end of every fitness correlation we report. It needs a deliberate strain-selection rule before those correlations carry weight.

## 6 Novel construction candidate — the one unmeasured pair

Of $\binom{10}{2} = 45$ pairs, **44 have a Costanzo DMF; 1 does not: YPL046C+YPL081W (ELC1+RPS9A)** — and it is also absent from Kuzmin 2018/2020. Missing is not the same as synthetic-lethal: an SL pair would show as a measured low fitness or an explicit flag, not a blank. A NaN just means the pair was never in a scored query×array orientation.

**SL check (SynthLethDB `Yeast_SL.csv`). not synthetic-lethal.** ELC1/YPL046C appears in **zero** SL records, and RPS9A/YPL081W is SL only with its paralog **RPS9B** — the classic redundant-duplicate case — not with ELC1.

**BioGRID / SGD second source.** Checked via SGD's aggregated interaction API: ELC1 has **97** recorded genetic-interaction partners and **RPS9A is not among them** — no interaction of any type (SL, negative or positive genetic, synthetic growth defect). ELC1 is well studied at 97 partners, so RPS9A's absence is a real gap, not sparse coverage.

**Recommendation. Construct it.** It is the one cell of the $10 \times 10$ matrix with no value in Costanzo, Kuzmin, SynthLethDB or BioGRID, and it is not synthetic-lethal, so it should be buildable. Tagged `novel` in the supplementary table.

# 7 Supplementary table — all 45 within-10 doubles

`experiments/010-kuzmin-tmi/results/construction_validation_doubles.csv` is the full supplementary table: **every** within-10 pair, with `tier` blank when not selected (`coverage` / `validation` / `novel` otherwise), Costanzo DMF ± SD, derived SE, $\varepsilon$, $p$, the **Kuzmin 2018/2020 DMF ± SD** columns, and the strain id. 45 rows = $\binom{10}{2}$; kept as a CSV rather than inlined because its roughly 15 columns are too wide to render at this page width.

# 8 What was actually built

Of the 14 doubles this selection called for, **13 were constructed**. **YKL033W-A × YJR060W (CBF1) failed to construct** (reported by Aurosish Sharma on handover of the 2026-08-06 plating). It is absent from the run-4 strain list, so the built panel is 12 singles + 13 doubles + WT. The missing pair is a construction failure, not a dropout to impute.

The plating that resulted is recorded in `experiments/W019-echo-crispr-array/data/run4_doubles_2026-08-06/PROVENANCE.md`: three independently randomized plates, WT at 28 wells and each mutant at 14 wells per plate, with both constituent singles of every double present on the same plate so $\varepsilon = f_{ab} - f_a f_b$ is computable within-plate.

# 9 Run 4 — the constructed panel, measured

The doubles selected above were built and plated on 2026-08-06 as three independently randomized 384-well plates (WT at 28 wells, every mutant at 14, 378 transfers + 6 blanks per plate), and imaged at an officially recorded **48 h 12 min** of incubation — not the nominal 48 h that the script filename abbreviates. The extra twelve minutes are common to every well on a plate, so they cancel in the per-plate normalization and cannot produce a strain-to-strain effect; the number is recorded precisely because it is the timing of record. Full provenance in `experiments/W019-echo-crispr-array/data/run4_doubles_2026-08-06/PROVENANCE.md`). Scoring is by `experiments/W019-echo-crispr-array/scripts/run4_doubles_48h.py`.

**No triple mutants.** This round carries **singles and doubles only** — the strain list has a `KO1` and a `KO2` column and 26 rows (WT, `s1`–`s12`, `d1`–`d13`). There is structurally no third-knockout field, so nothing here bears on trigenic interaction; that remains ahead of us.

**Registration.** All three plates resolved orientation confidently on strain structure and returned **6/6 blank wells empty**. That last number is load-bearing: this collaborator's Echo export *omits* the blank wells rather than listing them, so the blanks are reconstructed as the complement of the 378 transfers. Six of six empty means the reconstruction and the orientation are both right, checked against the image rather than assumed.

**Table 3** Run 4 per-plate quality control. All three plates pass. The 6/6 blanks matter independently: this Echo export omits the blank wells, so they are reconstructed as the complement of the 378 transfers, and six empty of six confirms both that reconstruction and the resolved orientation against the image.

| plate | occupied | blanks empty | WT CV | QC |
|---|---|---|---|---|
| P1 | 320/384 | 6/6 | 0.153 | pass |
| P2 | 345/384 | 6/6 | 0.158 | pass |
| P3 | 369/384 | 6/6 | 0.122 | pass |

## 9.1 Single-mutant fitness reproduces

Against the published Costanzo SMF the singles come back at **Pearson** $r = 0.706$ **(**$p = 0.015$**,** $n = 11$**)**. Eleven, not twelve, because this panel's `s7` is **YLR104W (LCL2)** where run 3 carried YPL056C (LCL1) — the two rounds' single panels differ by one strain, and the join is on the systematic ORF rather than on position in the list.

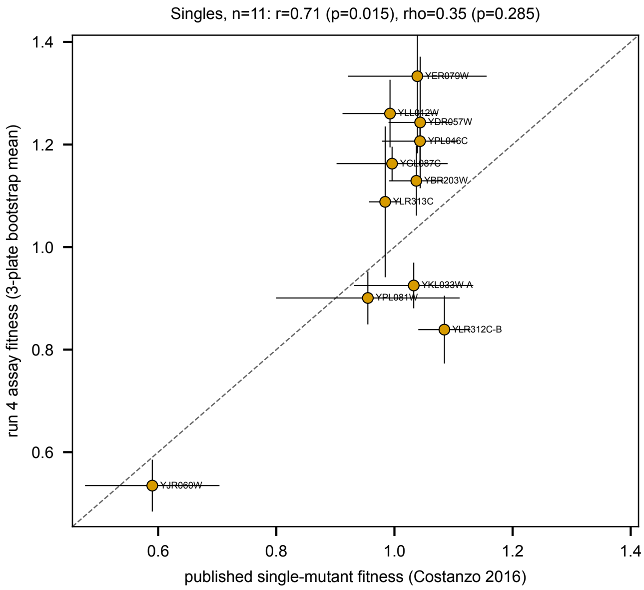


**Fig. 6** Run 4 single-mutant fitness (bootstrap mean over the three re-randomized plates, plate as the resampling unit) against published Costanzo 2016 SMF. Identity dashed; error bars are SE-vs-SE.

## 9.2 Singles and doubles on one axis

Both measurements come from the same three plates and the same code, so they belong on the same axis — and that is where the asymmetry shows. The singles track the identity line; the doubles sit low and flat, compressed into a narrow band regardless of what the reference says they should be.

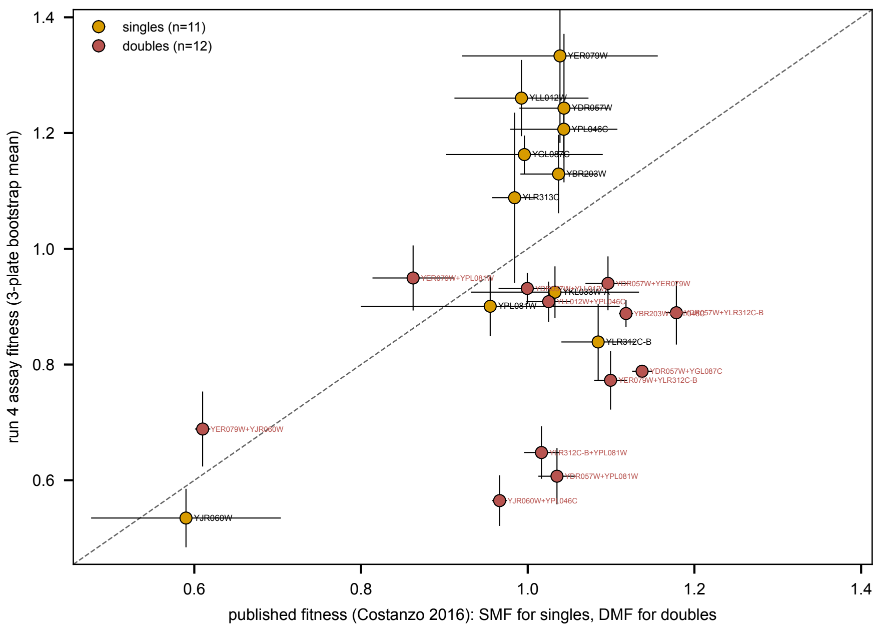


**Fig. 7** Run 4 against published Costanzo values on one axis: single mutants and double mutants, distinguished by colour. Identity dashed. The singles spread along the line; the doubles do not track it. **All four error bars are standard errors, but the two published ones are SEs of different provenance and this is worth stating.** For the *singles*, Costanzo's released "single mutant fitness stddev" column is, per the SOM, already a *bootstrap* SE over control screens — it is used as published and never divided by $\sqrt{n}$. For the *doubles*, the released value is a *sample SD over 4 colonies*, which we convert to an SE by dividing by $\sqrt{4}$ (verified: the ratio of the stored SD to the derived SE is exactly 2.000 for every pair). So one is an SE as published and the other is an SE we derived under the $n = 4$ determination recorded in `[[torchcell.datasets.scerevisiae.costanzo2016.noise-computation]]`. Vertical bars are ours, bootstrapped across the three plates.

### 9.3 Double-mutant fitness does not

The same comparison for the 13 constructed doubles gives $r = 0.255$ ($p = 0.42$, $n = 12$) — twelve because YPL046C+YPL081W, the novel pair, has no published DMF anywhere, which is why it was selected. Singles reproducing while doubles do not is the first sign of the problem the next section is about, and it is not a coincidence: the two are measured on the same plates by the same code.

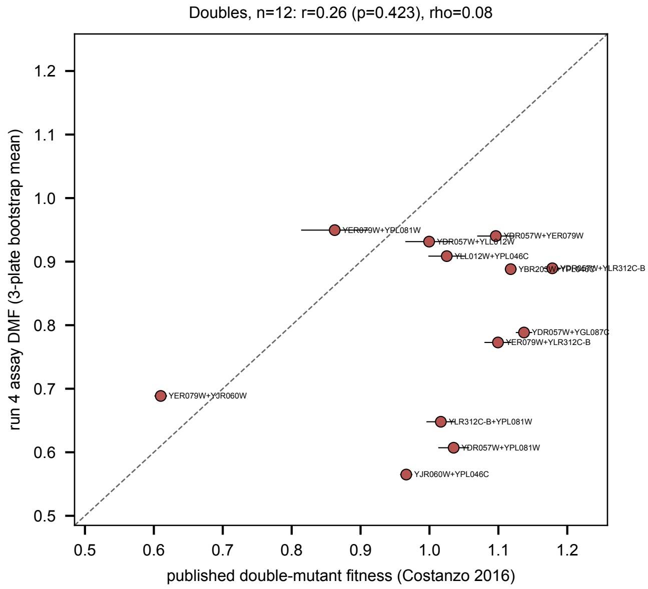


**Fig. 8** Run 4 double-mutant fitness against published Costanzo DMF. The 13th double, the novel YPL046C+YPL081W, is absent because no published value exists for it.

## 10 Digenic interaction versus Costanzo — measured, and not yet usable

The design supports the comparison cleanly. Both constituent singles of every double sit on the same plate, so

$$\varepsilon_{ab} \;=\; f_{ab} \;-\; f_a f_b \tag{1}$$

is formed entirely within a plate — no cross-plate normalization enters the interaction term, and the three plates give three independent replicates of the same quantity, which is what the across-plate bootstrap turns into an error bar.

Run against the published Costanzo $\varepsilon$ for the same pairs, the answer is $r = -0.245$ ($p = 0.44$, $n = 12$): no agreement, and the sign is wrong.

### 10.1 Every double, side by side

The full comparison, one row per constructed double, sorted by the published $\varepsilon$. The table is generated by `run4_doubles_48h.py` straight from `run4_doubles_vs_reference.csv` — no number here is transcribed by hand.

Read the $f_a f_b$ column first. It is the multiplicative expectation, and for ten of the thirteen pairs it exceeds 1.0 — which cannot be right for two deletion mutants and is what drives every $\varepsilon$ in the fourth column negative. YPL046C+YPL081W has no published $\varepsilon$ because it is the novel pair; it was constructed precisely because no reference value exists.

### 10.2 Our own significance — and which error term you use decides everything

Section 4.1 argued that a big-but-insignificant interaction is fixable because the error bar is set by how many colonies we plate, and that is our choice. This round is the first test of that argument, and the answer is more interesting than expected: **colony count was never the binding constraint.**

Two error terms can be put on $\varepsilon$, and the table reports both.

**Within-plate.** $\varepsilon$ is a function of three strain means measured on the same plate, so by the delta method

$$\mathrm{var}(\varepsilon) \;=\; \mathrm{var}(f_{ab}) \;+\; f_b^2\,\mathrm{var}(f_a) \;+\; f_a^2\,\mathrm{var}(f_b), \tag{2}$$

**Table 4** Every constructed double, sorted by published $\varepsilon$. **Bold marks Costanzo's intermediate-confidence tier**, $P < 0.05$ and $|\varepsilon| > 0.08$ — applied identically to both $\varepsilon$ columns, so ours is bolded from our $p$ and theirs from their published $p$, and the two are directly comparable. That tier is quoted verbatim from the Costanzo 2016 SI in the torchcell library mirror (`costanzoGlobalGeneticInteraction2016/si/si1.md`): "*We suggest three different thresholds [lenient ($P < 0.05$), intermediate ($P < 0.05$ and $|\varepsilon| > 0.08$), and stringent confidence ($P < 0.05$ and $\varepsilon > 0.16$ or $\varepsilon < -0.12$)] that strike different balances between false negatives and false positives.*" The intermediate tier is the one the paper itself works at — every genetic-interaction figure in it is drawn at $|\varepsilon| > 0.08$, $P < 0.05$ — and recomputing it over the reference table's 44 scored pairs reproduces the stored flag exactly. Read the $f_a f_b$ column first: the multiplicative expectation exceeds 1.0 for ten of thirteen pairs, which is what drives every $\varepsilon$ negative. $\mathrm{SE}_{\mathrm{within}}$ is colony scatter propagated by the delta method; $\mathrm{SE}_{\mathrm{across}}$ is the scatter of the three plate-level estimates; $p$ is a two-sided $t$-test on those three ($df = 2$). Generated by `run4_doubles_48h.py`; no value transcribed by hand. **Bold here means reproducible, not correct** — see the caveat below.

| | | | ours | | | | Costanzo |
|---|---|---|---|---|---|---|---|
| double | $f_{ab}$ | $f_a f_b$ | $\varepsilon$ | $\mathrm{SE}_{\mathrm{within}}$ | $\mathrm{SE}_{\mathrm{across}}$ | $p$ | $\varepsilon$ |
| YER079W+YPL081W | 0.950 | 1.224 | -0.274 | 0.046 | 0.195 | 0.295 | **-0.130** |
| YDR057W+YLL012W | 0.932 | 1.570 | -0.639 | 0.057 | 0.253 | 0.127 | -0.036 |
| YER079W+YLR312C-B | 0.773 | 1.092 | **-0.319** | 0.061 | 0.025 | 0.006 | -0.027 |
| YLR312C-B+YPL081W | 0.648 | 0.747 | -0.099 | 0.046 | 0.100 | 0.426 | -0.019 |
| YLL012W+YPL046C | 0.909 | 1.519 | **-0.611** | 0.062 | 0.108 | 0.030 | -0.010 |
| YER079W+YJR060W | 0.689 | 0.733 | -0.045 | 0.089 | 0.117 | 0.739 | -0.003 |
| YJR060W+YPL046C | 0.565 | 0.653 | -0.088 | 0.077 | 0.097 | 0.462 | +0.003 |
| YDR057W+YER079W | 0.940 | 1.712 | -0.772 | 0.054 | 0.454 | 0.231 | +0.012 |
| YBR203W+YPL046C | 0.888 | 1.372 | -0.483 | 0.063 | 0.149 | 0.083 | +0.036 |
| YDR057W+YPL081W | 0.607 | 1.139 | -0.532 | 0.047 | 0.275 | 0.193 | +0.039 |
| YDR057W+YLR312C-B | 0.890 | 1.023 | -0.133 | 0.058 | 0.085 | 0.257 | +0.047 |
| YDR057W+YGL087C | 0.788 | 1.442 | -0.654 | 0.056 | 0.170 | 0.061 | **+0.098** |
| YPL046C+YPL081W | 0.934 | 1.094 | -0.161 | 0.043 | 0.131 | 0.344 | — |

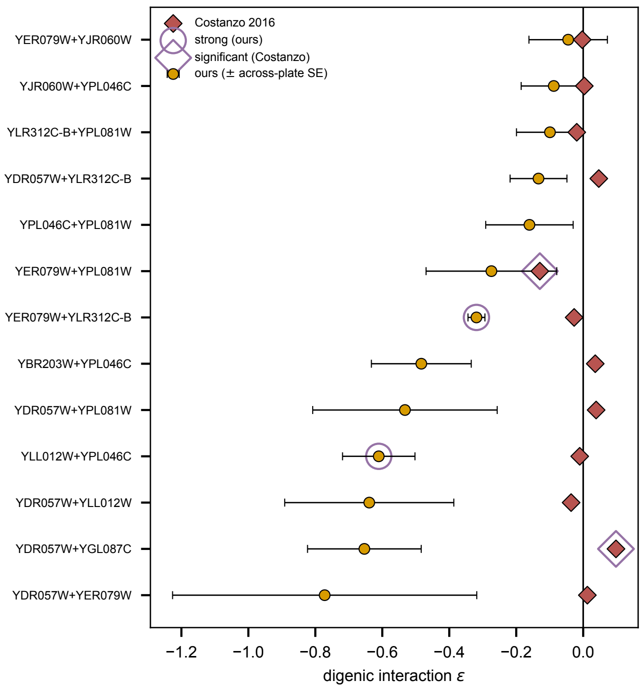


**Fig. 9** The same comparison per double, as intervals rather than points. Our $\varepsilon$ carries its across-plate SE; the Costanzo value is marked on the same row. The scatter of Fig. 10 answers whether the two agree overall; this answers, for each double, where our interval sits and where the published point falls relative to it. Rings mark the calls, using the same bar on both sides ($p < 0.05$ and $|\varepsilon| > 0.08$) — and they never coincide.

each variance being (colony SD$/\sqrt{n})^2$. This is the term Costanzo's own $p$-value is built from, and the one that 14 colonies per plate rather than 4 was supposed to shrink. It averages **0.058** across the thirteen doubles.

**Across-plate.** The scatter of the three per-plate $\varepsilon$ values, which carries the batch effect the within-plate term cannot see. It is **larger than the within-plate term for 12 of 13 doubles**, by a median factor of **2.4** and up to **8.5**.

That gap is the finding. Judged on colony scatter alone, **11 of 13** doubles would clear $p < 0.05$ (median $|\varepsilon|/\mathrm{SE} = 5.9$). Judged honestly, on three independent plate-level estimates, **2 of 13** do. Plating deeper does buy precision within a plate — it just buys precision on a quantity whose plate-to-plate reproducibility is the actual limit. The lever for this assay is *more plates*, not more colonies per plate, which is the opposite of what Section 4.1 assumed.

**Read the $p$ column with the caveat below.** The test is a two-sided $t$-test on three plate-level estimates, $df = 2$ — weak by construction, so a non-significant call here is poor evidence of absence. And more importantly: with the WT shift of Section 10.4 unresolved, this $p$ measures how *reproducibly* we recover a biased $\varepsilon$, not whether that $\varepsilon$ is right. Precision is not accuracy. The two doubles that do clear $p < 0.05$ (YER079W+YLR312C-B, YLL012W+YPL046C) are *not* the two Costanzo calls significant, and neither should be reported as a detected interaction.

## 10.3 Why this is an artifact and not a result

**It should not be read as a failure to reproduce Costanzo's interactions.** Four observations, all from `experiments/W019-echo-crispr-array/results/run4_interactions.csv`, identify it as a normalization problem upstream of Equation (1):

**Table 5** Four diagnostics identifying the interaction result as a normalization artifact upstream of Equation (1) rather than a failure to reproduce Costanzo.

| diagnostic | value |
|---|---|
| singles scoring *above* the on-plate WT | **8 of 12** (up to 1.333) |
| multiplicative expectation $f_a f_b > 1$ | **10 of 13** pairs (up to 1.712) |
| sign of $\varepsilon$ | **negative for all 13** |
| corr($\varepsilon$, $f_a f_b$) | $\mathbf{-0.926}$ ($p = 5 \times 10^{-6}$) |

A multiplicative null above 1.0 for two deletion mutants is not a meaningful expectation, and an $\varepsilon$ that correlates at $-0.93$ with its own expectation term is tracking the singles, not the doubles. Concretely, for the strongest published interaction in the panel:

**Table 6** The strongest published interaction in the panel, worked through. The $-0.654$ is arithmetic from two inflated singles, not a measured epistasis.

| YDR057W+YGL087C | $f_a$ | $f_b$ | |
|---|---|---|---|
| run 4 fitness | 1.243 | 1.163 | |
| expectation $f_a f_b$ | | | 1.446 |
| observed $f_{ab}$ | | | 0.788 |
| $\varepsilon$ (ours) | | | $\mathbf{-0.654}$ |
| $\varepsilon$ (Costanzo) | | | +0.098 |

That $-0.654$ is arithmetic from two inflated singles, not a measured epistasis.

## 10.4 Where the inflation comes from

The singles are on the same scale as run 3's — just shifted. Comparing our two rounds directly, the eleven strains common to both correlate at $r = 0.685$ ($p = 0.020$), but run 4 sits a near-uniform 1.26× above run 3 (median ratio; dividing it out leaves a mean absolute difference of 0.111). In run 3 every single scored *below* WT (0.67–0.92); in run 4 most score above it.

**Two correlations, not one.** Two similar numbers appear in this document and they are *different comparisons* on the same eleven strains — worth separating, because read together they look like one statistic quoted twice:

**Table 7** The two single-mutant correlations reported for run 4. Both are $n = 11$ over the same strains; they differ in what run 4 is compared *to*. Neither is a restatement of the other.

| comparison | against | $r$ | $p$ |
|---|---|---|---|
| run 4 singles vs run 3 singles | our own earlier round | 0.685 | 0.020 |
| run 4 singles vs Costanzo SMF | the published reference | 0.706 | 0.015 |

This matters for $\varepsilon$ and not for the correlations because $\varepsilon$ **is not scale-invariant**. Under a uniform inflation $f \to kf$,

$$\varepsilon' \;=\; kf_{ab} - k^2 f_a f_b, \tag{3}$$

so the expectation term grows quadratically while the observed term grows linearly, driving $\varepsilon$ negative in proportion to $f_a f_b$ — exactly the $-0.926$ correlation observed. A Pearson correlation, by contrast, is invariant to $k$, which is why the singles still reproduce the published values at $r = 0.706$ while every interaction is corrupted.

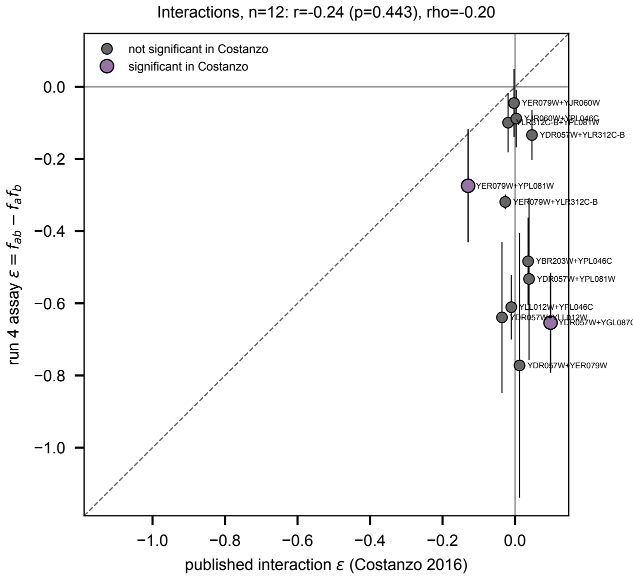


**Fig. 10** Run 4 $\varepsilon$ against published Costanzo $\varepsilon$, the two Costanzo-significant pairs marked. Every point sits below zero and the published spread is not recovered. This panel is published as the record of a diagnosed artifact, not as a measurement of epistasis.

## 10.5 What is not yet known

**Why the WT reference moved is untested.** At the colony level the WT median is *higher* than the mutant median on all three plates (ratio 0.90–0.94), so this is not simply sick or contaminated WT wells. Run 3 named its wild type `BY4741` explicitly; run 4's strain list gives it as `'-`. Whether that is a different stock, a different plating density, or something else has not been checked, and no correction should be applied before it is — rescaling to make the numbers look right would bury the question.

## 10.6 Diagnostic: it is not the wild-type wells

**This subsection reports a diagnostic, not a correction.** Re-scoring against the plate median rather than the WT wells does not make the numbers right — it makes them differently normalized, and the plate median is a poor stand-in for a wild type here because 25 of the 26 strains are mutants and 13 of those are doubles that genuinely grow less, so it compresses the very range the assay exists to measure. It was run to answer one narrow question: is the shift specific to the WT *wells*, or a property of the whole round? (`experiments/W019-echo-crispr-array/scripts/run4_wt_reference_diagnostic.py`.)

**Table 8** Swapping the normalization denominator. Every diagnostic is unchanged or slightly worse under the plate median, which rules the WT wells out as the explanation.

| | WT wells (as shipped) | plate median |
|---|---:|---:|
| singles above the reference | 8/12 | 9/12 |
| median run 4 / run 3 ratio | 1.263 | 1.368 |
| $f_a f_b > 1$ | 10/13 (max 1.712) | 10/13 (max 1.989) |
| $\varepsilon$ negative | 13/13 | 13/13 |
| corr($\varepsilon, f_a f_b$) | $-0.926$ | $-0.939$ |
| vs Costanzo SMF | $r = 0.706$ | $r = 0.708$ |

**The answer is no.** Every diagnostic is unchanged or marginally worse under the plate median. The inflation is not an artifact of the wild-type wells, so the hypothesis in the previous subsection — a different WT stock or plating density — is not supported and should not be pursued as though it were.

What the swap does clarify is where the asymmetry actually sits. It is not that the reference is low; it is that **the doubles are low relative to the singles**, and no choice of denominator changes that, because a common factor cancels from the ratio $f_{ab}/(f_a f_b)$ only when it applies to all three equally. Here the singles sit near 1.2 and the doubles near 0.8 against either reference.

## 10.7 Replicate structure — ours is technical, theirs is not

Two facts about how these numbers are made, which together explain most of the gap in precision and bound what our error bars can mean.

**Every strain is plated from a SINGLE picked colony.** All 14 wells of a strain on a plate, and all three plates, descend from one picked colony grown up and diluted. That makes the 42 colonies per strain **technical replicates of one picking event**, not independent biological samples. It is what makes the experiment tractable, but it has a consequence that must not be glossed: **the across-plate SE used throughout this section does not contain colony-picking variance.** Re-randomizing position across three plates averages out position and plate handling; it cannot average out a draw that was made once, upstream of all three. So the honest error bar of Section 10.2 is still an *under*-estimate with respect to the biological unit, and the $p$ values inherit that.

**Costanzo's single-mutant fitness is averaged over hundreds of screens.** From the SI (`costanzoGlobalGeneticInteraction2016/si/si1.md`, "Single mutant fitness standard"): "*Colony size measurements of SGA deletion and TS array mutant strains were based on an average of 350 replicate control screens ... query mutant strains were based on an average of 17 replicate control screens.*" Their SMF is not a 4-colony number — the 4 colonies apply to the double-mutant screens. Against 350 (or 17) independent control screens, three plates from one colony is a different order of replication, and that alone is enough to expect our agreement to be looser than their internal reproducibility.

**Not verified.** A reported reproducibility of roughly 0.9 between duplicate fitness measurements is plausible from this design but **could not be confirmed from our mirror**. The SI text describes fig. S2 as the reproducibility analysis and frames it in terms of precision/recall of *genetic interactions* across five replicate screens, not a fitness scatter; no correlation coefficient for duplicate fitness appears in the extracted text of either the paper or the SI. It may well sit inside a figure panel, which our OCR does not read. Treat the 0.9 as unconfirmed here rather than as absent.

## 10.8 A single-colony wild type, and what it can and cannot explain

The wild type is picked once per round like every other strain (Section 10.7), so the entire round's denominator rests on one colony. If that colony happened to be a weak grower, everything normalized to it is inflated by a constant — and because the three plates share it, no amount of within-round replication would reveal the problem. This is a much better candidate than the WT-wells hypothesis already rejected in Section 10.6, and it fits two observations directly: singles landing above 1.0, and run 4 sitting a near-uniform 1.26× above run 3, since the two rounds drew independent wild-type colonies.

**But it cannot be the whole story, and the arithmetic says so.** A reference error is a common factor $k$ applied to every strain on the plate, so it cancels out of any ratio between two strains:

**Table 9** A weak wild type rescales singles and doubles together, so the double-to-single ratio is invariant to it. That ratio is 0.758 where a multiplicative model predicts roughly the mean single itself (∼1.07); the discrepancy therefore survives any choice of reference.

| reference scale $k$ | mean single | mean double | double/single |
|---|---|---|---|
| 1.00 (as measured) | 1.066 | 0.809 | 0.758 |
| 1.26 | 1.344 | 1.019 | 0.758 |
| 1.50 | 1.600 | 1.213 | 0.758 |

So there are **two separable effects**, and conflating them is what made the earlier sections chase one explanation:

1. a *per-round scale* on the reference — consistent with a single-colony wild type, explains the 1.26× between rounds and the singles above 1.0, and does not touch $\varepsilon$'s *ordering*;
2. a *doubles-versus-singles* discrepancy — the double/single ratio is 0.758 where a multiplicative model wants ∼1.07 — which no reference choice can produce and which is what actually corrupts $\varepsilon$.

Effect (1) now has a plausible mechanism. Effect (2) does not, and it is the one that matters for interactions.

### 10.9 What is not yet known

The WT-wells explanation is tested and rejected (Section 10.6); a single-colony wild type plausibly accounts for the per-round scale but provably not for the doubles-versus-singles gap (Section 10.8). What remains open is that second effect: why the doubles sit low relative to the singles under any normalization. It may be partly real biology at 48 h 12 min on this medium; nothing here tests that.

What this round *cannot* distinguish is a per-strain construction or picking effect from a genetic one, because each strain enters the experiment through a single picked colony (Section 10.7). A double whose founding colony happened to be a poor grower is indistinguishable, in these data, from a double with a real negative interaction — and with all three plates sharing that colony, no amount of within-round replication separates them. **The discriminating experiment is independent pickings**: two or more colonies per strain, carried as separate lineages through the whole round. That converts the replicate unit from technical to biological and is the same change that would let the across-plate SE of Section 10.2 mean what it currently only appears to mean.

Until then, no $\varepsilon$ from this round should be reported as an interaction, and no rescaling should be applied to make the numbers look reasonable: every correction available here is cosmetic and would bury the question.

## 11 Detection overlays — what was actually measured

Every number in Sections 9–10 is a colony area read off these three images. The overlays are the audit trail: each detected colony's boundary is drawn on the original plate photograph, coloured by whether the pipeline accepted it, and the fitted grid nodes are marked with cyan crosses so a node sitting off its colony stays visible rather than silently mis-assigning a strain.

Plate addresses run down both sides (rows A–P) and across top and bottom (columns 1–24), set in the outside margin rather than on the agar — drawn in-image they collide with edge-row colonies, and no amount of halo makes small glyphs legible over agar, plastic and the dark lid at once. The headers are *plate* addresses under the resolved orientation, not raw image indices, so a well read off the figure can be looked up directly in the picklist.

**Table 10** Detection overlay colour key. Only in-gel, on-grid instances are drawn, so off-plate and frame detections never appear.

| outline | meaning |
|---|---|
| green | accepted – in gel, on grid, circular, single colony |
| red | multiple colonies detected on one well; rejected |
| orange | neighbour of a multi-colony well |
| purple | non-circular |
| blue | recovered faint colony, or straddling the gel edge |
| cyan cross | fitted grid node |

**One thing to look at.** Row P carries a visibly higher share of non-green outlines than the interior rows on all three plates, and occupancy differs substantially between plates (320 / 345 / 369). Whether the bottom row is subject to an edge effect in this rig has *not* been tested; it is a separate question from the wild-type shift of Section 10.4 and is noted here only because these figures make it visible.

## 12 Related notes

- `[[experiments.010-kuzmin-tmi.scripts.topk_triples_from_constructed_10]]`
- `[[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]`
- `[[experiments.010-kuzmin-tmi.scripts.constructed_10_dmf_reference]]`
- `[[experiments.010-kuzmin-tmi.12_panel_crispr_fitness_assay]]` — the assay these doubles validate

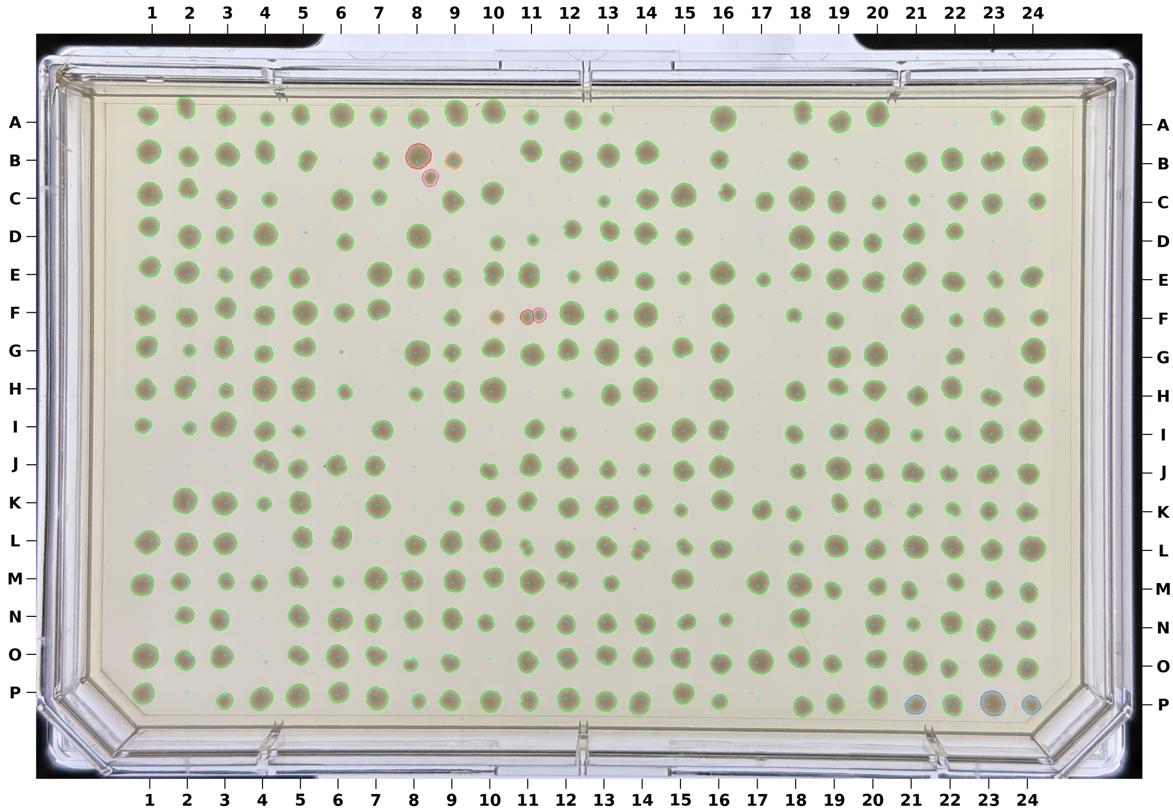

**Fig. 11** Run 4, plate P1 (5 nL, OD 1.0, 48 h 12 min). 320 of 384 wells occupied, 6/6 blanks empty, WT CV 0.153.

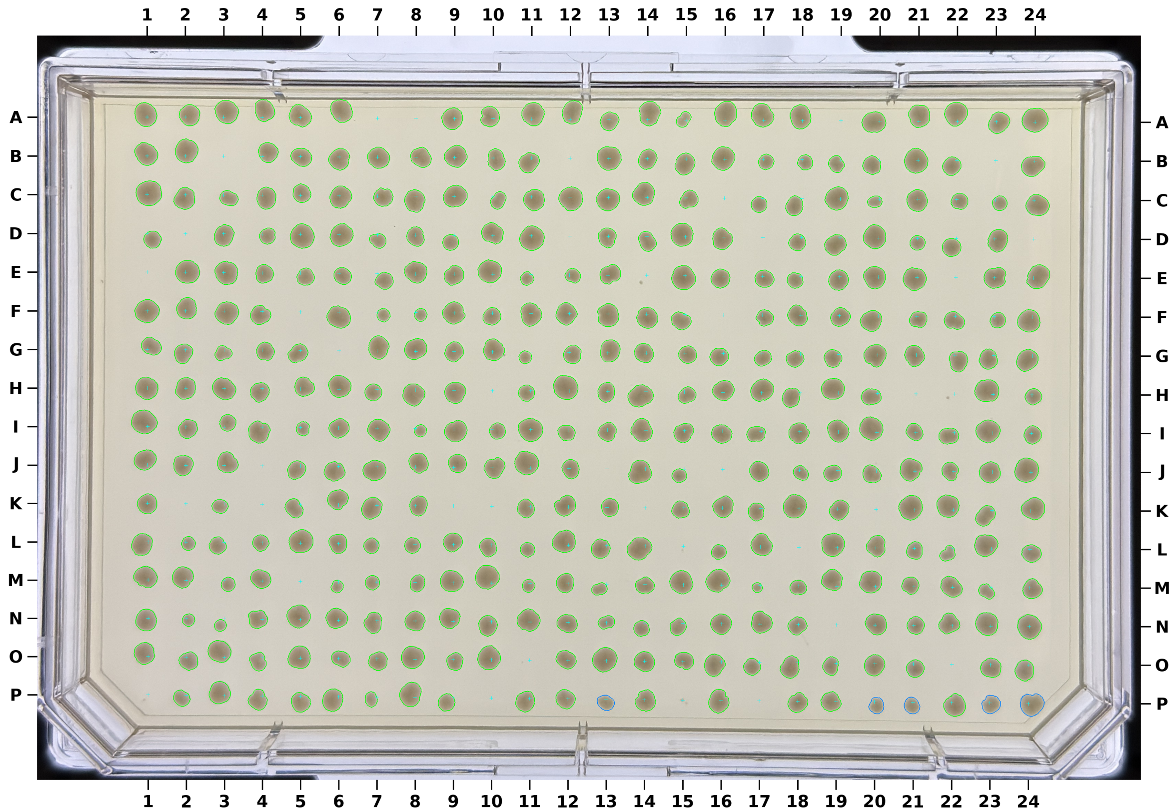

**Fig. 12** Run 4, plate P2 (48 h 12 min). 345 of 384 occupied, 6/6 blanks empty, WT CV 0.158.

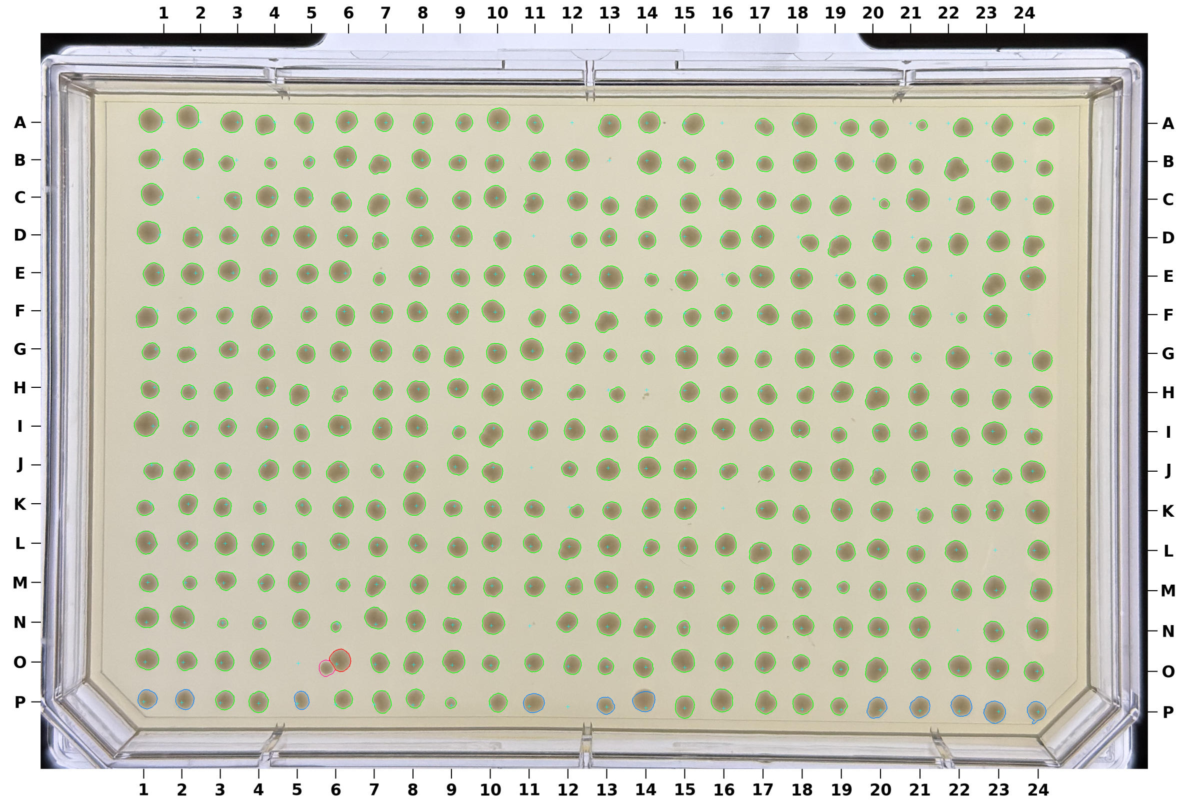


**Fig. 13** Run 4, plate P3 (48 h 12 min), the cleanest of the three. 369 of 384 occupied, 6/6 blanks empty, WT CV 0.122.